

Land Use / Land Cover (LULC) Classification

Experiment 3: Effect of Increased Sample Size Using Raw Spectral Bands

Random Forest vs. XGBoost — Effect of Dataset Size on Raw-Band Classification

Project Report

1. Objective

The objective of this experiment was to investigate the effect of increasing the number of samples per class on LULC classification performance while maintaining a simple feature representation based only on raw Sentinel-2 spectral bands.

In this experiment, the number of samples per class was increased to 10,000, resulting in a total dataset of 90,000 samples across nine LULC classes. Both Random Forest and XGBoost classifiers were evaluated using the same feature configuration.

2. Dataset and Feature Configuration

Parameter	Configuration
Experiment ID	EXP-03
Experiment Type	Sampling / Dataset Size Experiment
Input Data	Raw Sentinel-2 multispectral data
Temporal Coverage	Single seasonal observation
Spectral Bands	B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12
Total Input Features	10
Spectral Indices	None
Additional Feature Engineering	None
Number of LULC Classes	9
Samples per Class	10,000
Total Samples	90,000
Test Samples	18,000
Test Set Size	20%
Class Labels	10, 20, 30, 40, 50, 60, 80, 90, 95
Models Evaluated	Random Forest, XGBoost

3. Class Distribution

The dataset was perfectly balanced, with 10,000 samples for each of the nine LULC classes.

Class	Samples
10	10,000
20	10,000
30	10,000
40	10,000

Class	Samples
50	10,000
60	10,000
80	10,000
90	10,000
95	10,000
Total	90,000

4. Models Evaluated

Two ensemble machine learning classifiers were evaluated:

- Random Forest Classifier
- XGBoost Classifier

Both models used the same 10 raw spectral features and the same dataset.

5. Overall Performance

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Random Forest	85.00%	0.85	0.85	0.85	0.84
XGBoost	84.74%	0.85	0.85	0.85	0.85

Best Performing Model

Random Forest achieved the highest accuracy:

- Accuracy: 85.00%
- Macro F1-score: 0.85

XGBoost achieved 84.74% accuracy, which was approximately 0.26 percentage points lower than Random Forest.

6. Random Forest Class-wise Performance

Class	Precision	Recall	F1-score	Support
10	0.79	0.79	0.79	2,011
20	0.79	0.82	0.80	1,934
30	0.75	0.75	0.75	1,987
40	0.86	0.88	0.87	1,973

Class	Precision	Recall	F1-score	Support
50	0.84	0.80	0.82	2,104
60	0.86	0.85	0.86	2,027
80	0.98	0.99	0.99	1,979
90	0.84	0.80	0.82	2,101
95	0.89	0.94	0.91	1,884
Macro Avg	0.85	0.85	0.85	18,000
Weighted Avg	0.84	0.85	0.84	18,000

7. XGBoost Class-wise Performance

Class	Precision	Recall	F1-score	Support
10	0.79	0.79	0.79	2,000
20	0.80	0.80	0.80	2,000
30	0.76	0.75	0.75	2,000
40	0.88	0.87	0.88	2,000
50	0.83	0.84	0.83	2,000
60	0.85	0.87	0.86	2,000
80	0.99	0.98	0.98	2,000
90	0.80	0.85	0.82	2,000
95	0.95	0.88	0.92	2,000
Macro Avg	0.85	0.85	0.85	18,000
Weighted Avg	0.85	0.85	0.85	18,000

8. Key Observations

1. Increasing sample size did not improve accuracy

- The dataset was increased substantially to 90,000 samples, but Random Forest still achieved approximately 85% accuracy.
- This indicates that simply increasing the number of samples did not produce a meaningful improvement under the given feature configuration.

2. Simple spectral representation remained the limiting factor

- Only 10 raw spectral bands were used in this experiment. No temporal information, spectral indices, or additional feature engineering was incorporated.

- The results suggest that the additional samples alone were insufficient to capture the class differences that could potentially be represented through richer features.

3. Random Forest slightly outperformed XGBoost

- Random Forest achieved 85.00% accuracy, while XGBoost achieved 84.74% accuracy.
- The difference was small, indicating comparable performance between the two models for this configuration.

4. Class 80 remained highly separable

- Both models achieved approximately 0.98–0.99 F1-score for Class 80, indicating strong spectral separability for this class.

5. Class 30 remained the most challenging

- Both models achieved approximately 0.75 F1-score for Class 30, suggesting greater spectral overlap with other LULC categories.

9. Comparison with Experiment 1

Because both experiments use nine classes and raw spectral bands, this provides a useful comparison:

Experiment	Features	Samples	Classes	RF Accuracy	XGB Accuracy
EXP-01	40	27,000	9	85.00%	87.02%
EXP-03	10	90,000	9	85.00%	84.74%

The number of samples increased from 27,000 to 90,000, but Random Forest accuracy remained at 85%.

This provides experimental evidence that, under these configurations, increasing sample quantity alone did not lead to an observable improvement in performance.

At the same time, the comparison should be interpreted carefully because EXP-01 contains four-season information, whereas EXP-03 uses a single seasonal observation.

10. Conclusion

Experiment 3 investigated the effect of increasing the dataset size while maintaining a simple raw-band feature representation. A balanced dataset of 90,000 samples was created, with 10,000 samples for each of the nine LULC classes.

Ten raw Sentinel-2 spectral bands were used as input features without spectral indices or additional feature engineering. Random Forest achieved an accuracy of 85.00%, while XGBoost achieved 84.74%.

The results indicate that increasing the number of samples alone did not produce a significant improvement in classification performance under this configuration. The experiment further highlights the potential importance of incorporating richer feature representations, including multi-seasonal information and derived spectral features, for improving LULC classification performance.